

Medical Aid Customer Review Analysis

Pipelines and Text Data: LDA Topic Modelling & Supervised Sentiment Classification

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1. Executive Summary

A South African medical aid scheme asked two questions about its customer reviews: what are the broad areas of concern, and what is the overall sentiment? Using a public insurance-review corpus as a proxy (Rihab, 2021), this analysis applied eight-topic LDA for concern discovery and a tuned TF-IDF plus Logistic Regression pipeline for binary sentiment classification. Policy, contact and communication problems emerged as the dominant concern, carrying a 23.9 percent negative rate, far ahead of auto claims (13.6 percent) and jewelry claims (9.3 percent). The sentiment model detects about 83 percent of negative reviews (macro-F1 0.798 on the held-out test set). The top recommendation is to reduce response times and improve contactability on policy and membership queries, the area that drives the most dissatisfaction by both rate and volume.

2. Introduction and Client Problem

A South African medical aid scheme faces rising negative public reviews on platforms such as Hello Peter. Management needs data-driven insight into two questions: what broad areas drive customer concern, and what is the overall sentiment of the customer base?

This report delivers both. Latent Dirichlet Allocation (LDA) topic modelling surfaces recurring concern themes from review text, while a supervised TF-IDF plus classifier pipeline classifies customer sentiment. LDA suits this task because it represents each review as a mixture of topics and each topic as a distribution over words (Blei, Ng and Jordan, 2003), capturing short reviews that may raise several concerns at once. A supervised sentiment pipeline was chosen over VADER alone because it learns from the data, applies to unseen reviews from the same source, and provides interpretable feature coefficients that VADER cannot.

Three success criteria guide the work: interpretable concern topics with business-meaningful labels; a sentiment model that prioritises recall on negative reviews (because a missed dissatisfied customer is the costlier error); and a cross-analysis linking concern topics to negative sentiment so the client can prioritise remedial action.

3. Dataset and Suitability

The dataset is "Reviews about Insurance Companies" published on Kaggle (Rihab, 2021). A medical aid is a health-insurance product, and these reviews cover the same service touchpoints targeted by the brief: sign-up and onboarding, agent and call-centre interactions, claims processing, and coverage decisions. The proxy is therefore appropriate.

The raw dataset contains 44,934 rows and 4 columns (Unnamed: 0, Customer Name, Customer Location, Review Text) with zero missing values and no exact duplicate rows. 247 duplicate review texts were removed, leaving 44,687 unique reviews. After preprocessing and a minimum-length filter (fewer than 3 cleaned tokens), the final modelling corpus is 43,830 reviews.

No rating, score, or label column exists, so supervised sentiment ground-truth labels are unavailable. Sentiment labels are derived using VADER, a lexicon-and-rule model designed for short, informal review

text (Hutto and Gilbert, 2014). The limitation is plain: sentiment metrics measure agreement with lexicon-derived labels rather than human ground truth. This is mitigated by manual validation and by prioritising model interpretability.

The corpus is not South African and not medical-aid specific, so findings transfer by analogy only (proxy limitation). Public reviews over-represent strong opinions and this corpus skews positive. Topics were therefore discovered on the full cleaned corpus, then cross-analysed by negative sentiment rate to surface the concern areas tied to dissatisfaction. VADER can misread sarcasm and mixed sentiment. The Unnamed: 0 column (a leftover index) was dropped, Customer Name was removed for the ethics requirement, and Customer Location (mostly US at about 90 percent) was retained for descriptive EDA only.

4. Analysis Plan

The analysis follows a staged pipeline from cleaning through to cross-analysis.

Stage	Action	Output
Data cleaning	Drop index and name columns, remove 247 duplicate review texts, normalise whitespace	44,687 clean reviews
Text EDA	Review-length distribution, top unigrams and bigrams, location breakdown	Interpreted visuals
Preprocessing	Lowercase, strip URLs/HTML/punctuation, remove stopwords (retaining negation), lemmatise, drop reviews with < 3 tokens	43,830 reviews with clean_text column
Sentiment labelling	Apply VADER compound score with standard thresholds (Hutto and Gilbert, 2014), validate a manual sample	Labelled dataset
Feature extraction	CountVectorizer for LDA, TfidfVectorizer for sentiment	Document-term matrices
Topic modelling	Train LDA over k = 4 to 14, select by c_v coherence, label topics	8 labelled topics
Sentiment modelling	Stratified split, compare MNB / LR / LinearSVC, tune best with GridSearchCV	Tuned Logistic Regression pipeline
Evaluation	Classification report, confusion matrix, coefficient analysis, error analysis	Metric tables
Cross-analysis	Negative sentiment rate by topic, complaint vocabulary in strongly negative reviews	Concern-priority ranking

5. Exploratory Data Analysis

After dropping the redundant index and the Customer Name column (for the ethics requirement), normalising whitespace, and removing 247 duplicate review texts, the cleaned corpus is 44,687 unique reviews with zero missing values. The reviews are treated as English for the NLP pipeline; no automated language detection was run.

Reviews are short and right-skewed. The median is 19 words (109 characters) and the mean 27.5 words (156.6 characters). A single-word minimum and a long tail (maximum 866 words, 75th percentile 34 words) are present. A minimum-length filter is applied in preprocessing. The small tail beyond the 99th percentile needs no truncation.

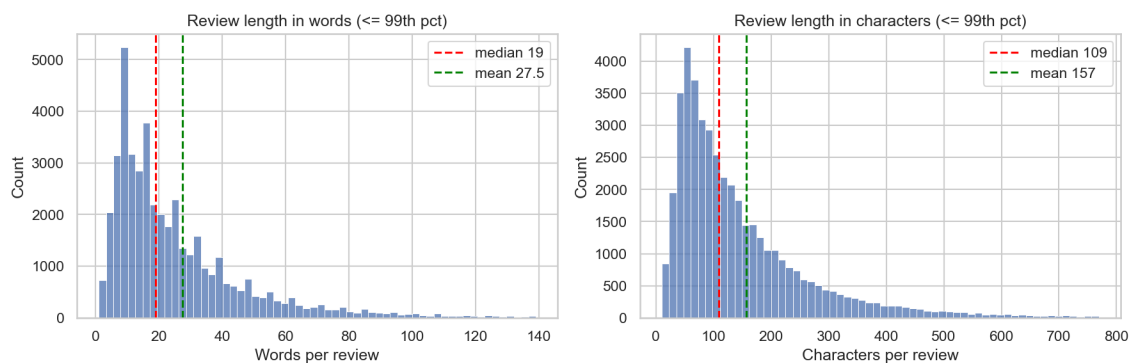


Figure 1: Review length distribution (words and characters, capped at 99th percentile).

The top unigrams after stopword removal are domain terms (insurance, service, process, policy, customer, coverage) and positive evaluative terms (easy, great, helpful, professional, excellent). The corpus skews positive. Bigrams confirm this: the most frequent is customer service (3,814), followed by process and onboarding phrases and evaluative bigrams such as "highly recommend". The brand name jewelers mutual (1,440) signals that specific insurer names are present.

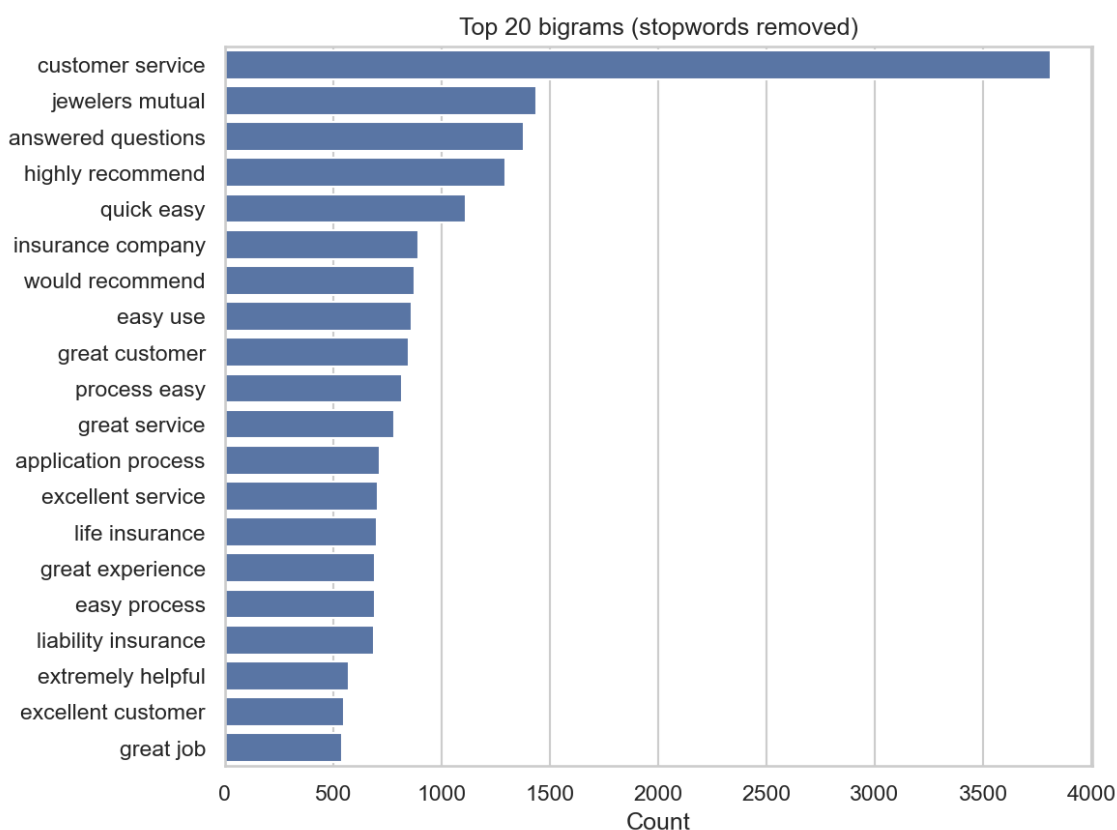


Figure 2: Top 20 bigrams after stopword removal.

About 90 percent of reviews are recorded as US-based. CR and BA are the next-largest location codes, retained for descriptive EDA only and never used as model features. The corpus is US-centric and not South African, which is the core proxy limitation.

6. Methodology

The preprocessing function lowercases each review, strips URLs and HTML tags, retains only alphabetic tokens, removes English stopwords (Bird, Klein and Loper, 2009), and lemmatises each remaining token. Negation words (not, no, never, cannot, and others) are excluded from the stopwords list because they invert the sentiment of adjacent terms. The original Review Text column is preserved unchanged for VADER labelling, which relies on capitalisation and punctuation cues (Hutto and Gilbert, 2014). Reviews reducing to fewer than 3 cleaned tokens are dropped, removing 857 reviews and leaving 43,830 for modelling.

LDA uses a CountVectorizer (min_df=20, max_df=0.5) because LDA models raw word counts, not TF-IDF weights (scikit-learn developers, 2024). The sentiment pipeline uses TfidfVectorizer with unigrams and bigrams so that negation phrases such as 'not helpful' are captured as features. The data is split 80/20 with stratification on the binary target. TF-IDF vectorisation happens inside the pipeline to prevent information leakage from test to train.

LDA was preferred over NMF and BERTopic. NMF yields comparable themes but no probabilistic document-topic mixture, and BERTopic adds transformer dependencies that this lightweight, reproducible

setting does not warrant. The classifier choice follows the same fit-to-purpose logic: Logistic Regression over Multinomial Naive Bayes and Linear SVC, for stronger negative recall and interpretable coefficients.

The analysis ran in Python. Data handling used pandas (pandas development team, 2024) and NumPy (Harris et al., 2020); the figures were produced with Matplotlib (Hunter, 2007) and seaborn (Waskom, 2021).

7. Areas of Concern: LDA Topic Modelling

LDA was fit for $k = 4, 6, 8, 10, 12, 14$ topics and each scored by c_v coherence (Rehurek and Sojka, 2010) and perplexity. Coherence peaks at $k = 8$ ($c_v = 0.5816$). Perplexity increases monotonically with k and diverges from coherence, as expected: perplexity measures held-out fit rather than human interpretability, and the two are known to disagree (Chang et al., 2009). Coherence and interpretability are prioritised. Eight topics are selected.

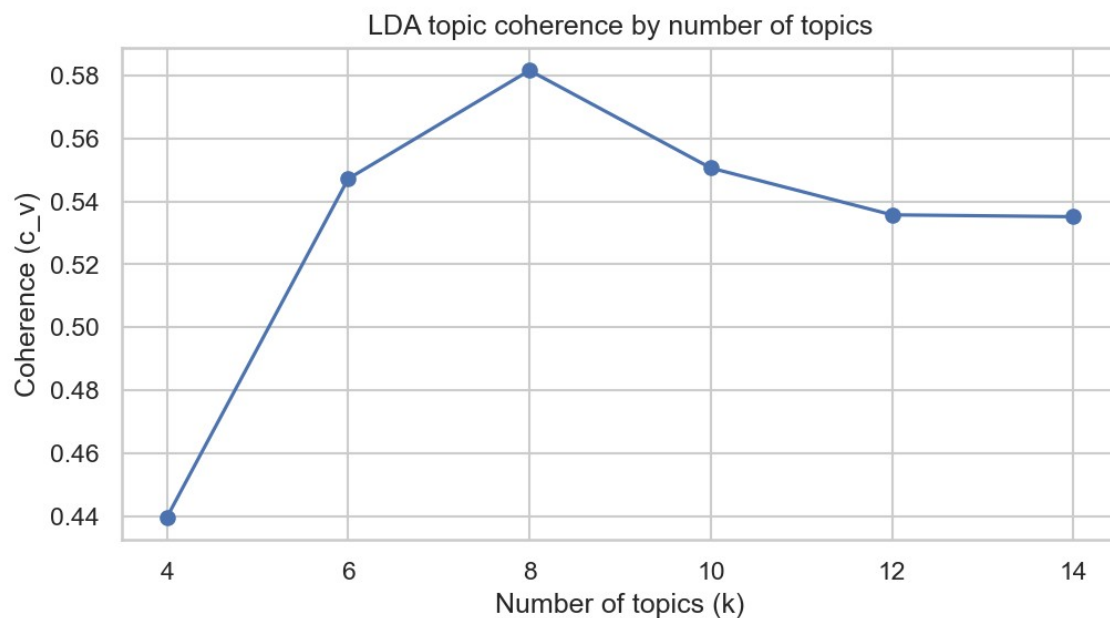


Figure 3: LDA topic coherence (c_v) by number of topics.

The final model is refit with `learning_method='batch'` and 25 iterations for a stable solution (Blei, Ng and Jordan, 2003; scikit-learn developers, 2024). The eight labelled topics are:

Topic	Top Words (summary)	Label	Group
0	recommend, great, excellent, best, service	General satisfaction and recommendation	Praise
1	easy, quick, online, website, fast, price	Ease and speed of the online process	Process
2	plan, medicare, health, coverage, option	Plan and coverage selection	Process
3	helpful, professional, knowledgeable, representative	Agent helpfulness and professionalism	Praise

Topic	Top Words (summary)	Label	Group
4	claim, car, repair, vehicle, njm	Auto claims handling	Claims
5	not, no, policy, call, email, time	Policy, contact and communication problems	Concern
6	application, policygenius, step, life, bond	Application and onboarding process	Process
7	jeweler, jewelry, ring, mutual, claim	Jewelry insurance and claims (Jewelers Mutual)	Product/claims

By prevalence, praise and ease themes lead (Topic 3: 9,076 reviews; Topic 1: 7,326; Topic 0: 6,500), consistent with the positive skew. The leading concern theme (Topic 5: 6,374 reviews) is the fourth most prevalent, showing that policy and communication problems are a substantial minority.

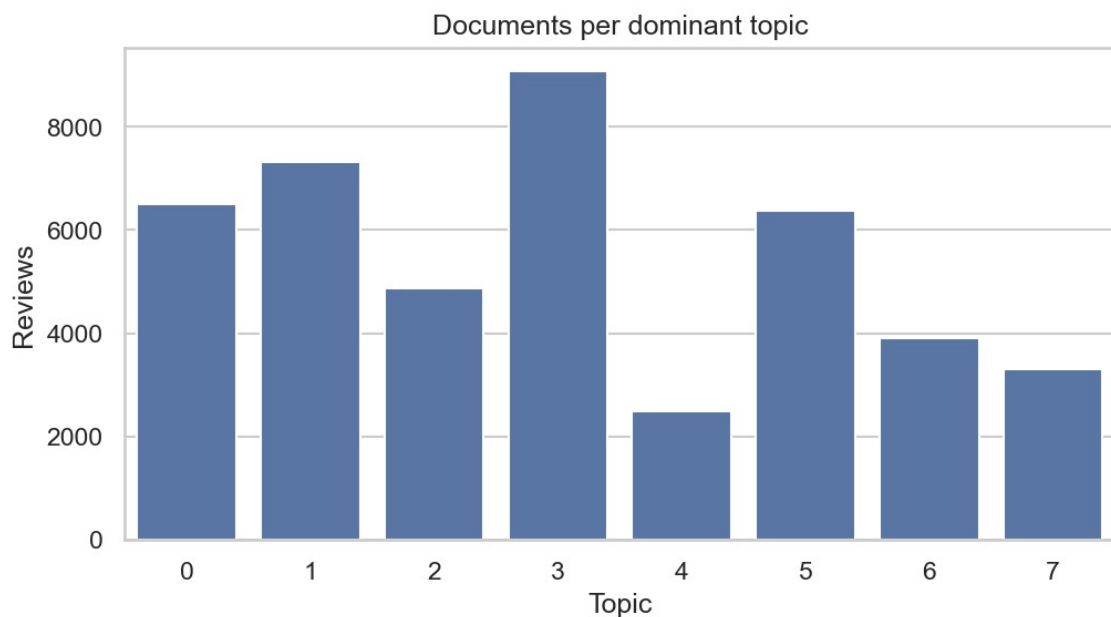


Figure 4: Document count per dominant topic.

8. Customer Sentiment: Classification

VADER is applied to the original Review Text (not the cleaned text) to preserve capitalisation and punctuation cues (Hutto and Gilbert, 2014). The compound score maps to three classes using standard thresholds: compound ≥ 0.05 positive, ≤ -0.05 negative, else neutral. The distribution is: positive 88.6 percent (38,812), negative 6.1 percent (2,674), neutral 5.3 percent (2,344).

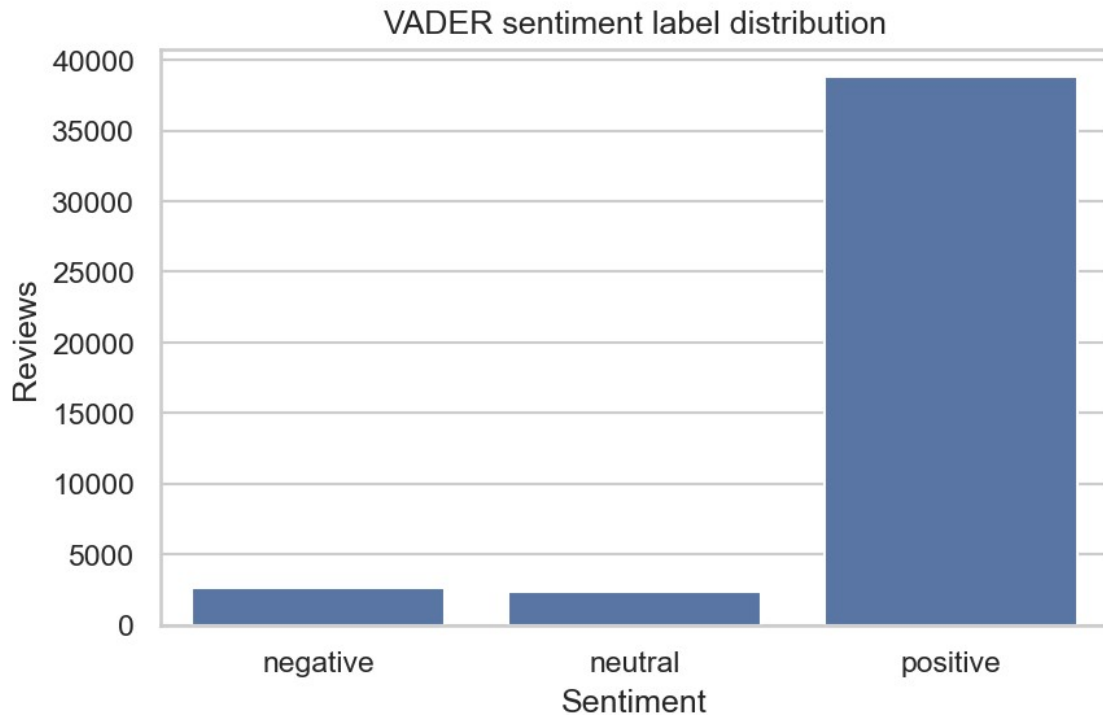


Figure 5: VADER three-class sentiment distribution.

Manual validation of five reviews per class shows VADER's positive class is reliable but the negative and neutral classes are noisy. Several neutral reviews are mildly positive (praise vocabulary outside VADER's lexicon), and some positive reviews are scored negative where insurance terms such as claim, loss and struggled carry negative lexicon weight. The supervised model therefore uses a binary target: negative versus non-negative. Merging neutral into non-negative absorbs the boundary ambiguity. The binary negative share is 6.1 percent (2,674 of 43,830).

Three linear classifiers suited to high-dimensional sparse text are compared with stratified 5-fold cross-validation on the training set (Pedregosa et al., 2011; scikit-learn developers, 2024). LinearSVC achieves the best macro-F1 (0.795) but Logistic Regression achieves the best negative recall (0.839 versus 0.699). Multinomial Naive Bayes is poor on the negative class (recall 0.034) because, without class weighting, it defaults to the majority class. Logistic Regression is taken forward: the client's priority is detecting dissatisfaction, where negative recall matters most. Macro-F1 is reported alongside it because it weights both classes equally under the class imbalance. It also provides probability estimates for threshold control and interpretable coefficients.

GridSearchCV tunes the Logistic Regression pipeline over ngram range, minimum document frequency, and regularisation strength C. The best configuration selects bigrams (ngram_range (1,2)), min_df=3, and C=2.0. The tuned cross-validation macro-F1 is 0.783, up from 0.768 untuned.

9. Model Evaluation

On the held-out test set the tuned model reaches a macro-F1 of 0.798 and accuracy of 0.940. For the negative class (the focus), precision is 0.504, recall is 0.834, and F1 is 0.628. The confusion matrix shows

446 of the 535 true negative reviews are caught and 89 are missed, while 439 non-negative reviews are flagged as negative.

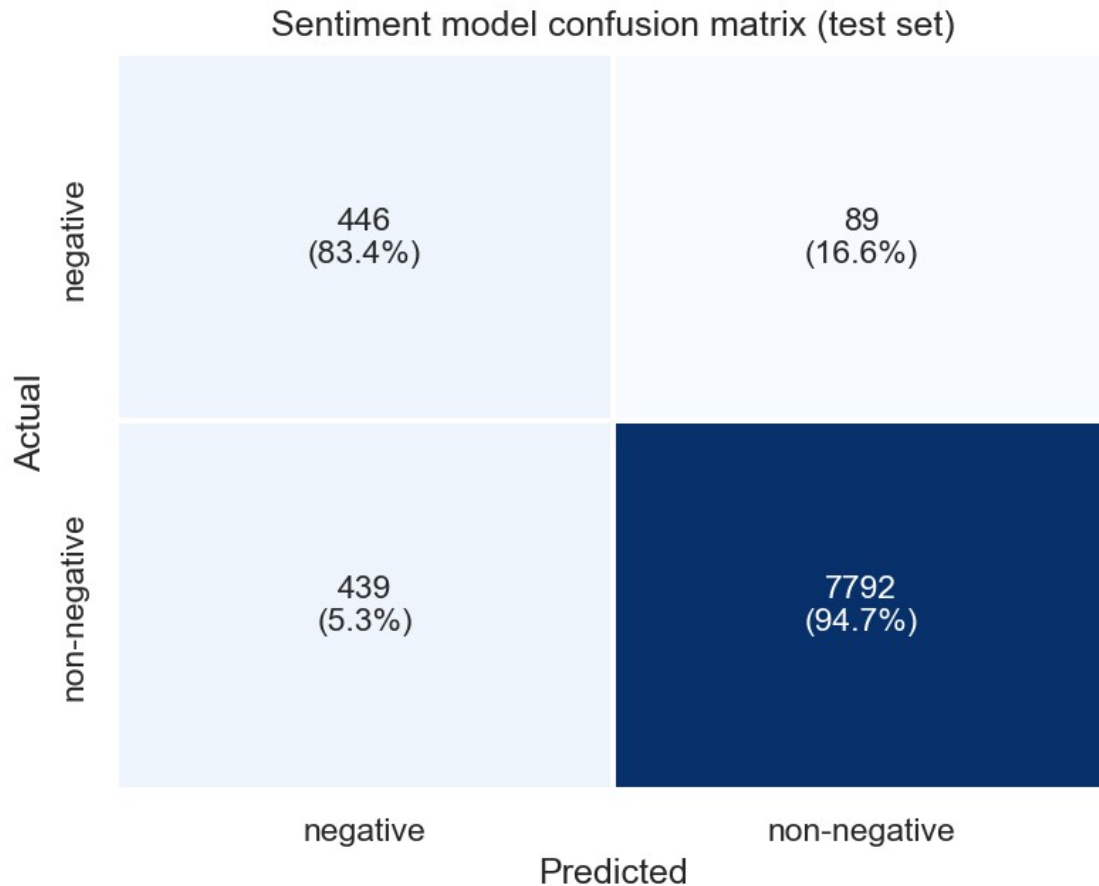


Figure 6: Sentiment model confusion matrix (test set).

In client terms the model detects about 83 percent of dissatisfied customers. This suits a screening tool whose main risk is missing dissatisfaction. Roughly half of the reviews it flags are false alarms, so flagged reviews are best treated as a prioritised queue for human review rather than a final verdict.

The Logistic Regression coefficients show which words drive each prediction. Terms pushing toward negative are genuine dissatisfaction vocabulary: negations (no, nothing, not), problem words (difficult, problem, wrong, hard, confusing), emotional terms (frustrating, terrible, disappointed, stress), and domain-specific negatives (accident, damage). Terms pushing toward non-negative are praise vocabulary (easy, great, helpful, excellent, best, efficient, awesome, good, happy, courteous). The model learned linguistically coherent drivers, giving confidence it captures real sentiment language.

Error analysis reveals two patterns. Some errors are genuine model mistakes: short or subtly worded complaints, or positive reviews containing negatively weighted domain words. The more revealing pattern is that many apparent errors are cases where the VADER label is itself wrong and the model's prediction is more accurate. True performance against human judgement may be somewhat better, but it cannot be confirmed without a human-labelled sample. This reinforces the lexicon-derived label limitation and supports using the model as a screening tool whose flags are reviewed by a person.

10. Cross-Analysis: Which Concerns Drive Negativity

Three topics carry almost all the dissatisfaction. Policy, contact and communication problems is the highest at 23.9 percent negative, ahead of auto claims handling (13.6 percent) and jewelry insurance and claims (9.3 percent). The remaining topics are overwhelmingly positive. Agent helpfulness is the lowest at 0.8 percent.

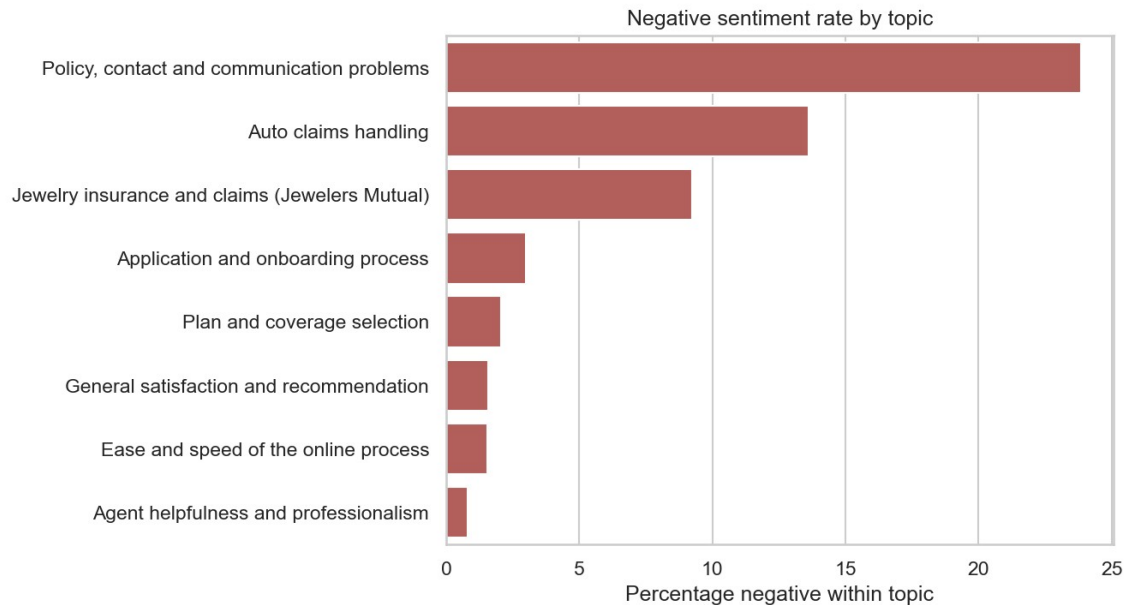


Figure 7: Negative sentiment rate by topic.

This reranks the topics very differently from raw prevalence. Agent helpfulness was the most prevalent topic (9,076 reviews) yet is the least negative, making it a clear strength. Policy, contact and communication problems was only fourth by prevalence (6,374 reviews) but is the most negative, making it the genuine priority. Combining rate and volume, policy and communication problems produce the largest absolute number of negative reviews (roughly 1,500), well ahead of auto claims (about 340) and jewelry claims (about 310).

The strongly negative subset (1,725 reviews with VADER compound ≤ -0.3) is dominated by negations (not 1,139, no 721) and contact and policy terms: policy, time, company, call, email, problem, service, process. Specific words sharpen the picture: 'call', 'email', 'time' and 'day' point to difficulty reaching the company and slow responses, while 'policy' and 'problem' point to unresolved policy and account issues.

11. Recommendations

The findings translate into four priorities for the medical aid, read through the insurance-review proxy.

Priority	Recommendation	Evidence
1	Fix policy servicing and communication first	Highest negative rate (23.9%) and largest absolute count (~1,500 negative reviews). Complaint terms: call, email, time, problem.

Priority	Recommendation	Evidence
2	Strengthen claims handling	Second tier: auto claims 13.6% negative, jewelry claims 9.3%. Complaints around turnaround and disputed valuations.
3	Protect and replicate agent and onboarding strengths	Agent helpfulness (0.8% negative) and easy online process (1.6%) are clear strengths. Praise vocabulary: helpful, professional, knowledgeable, easy.
4	Deploy the sentiment model as topic-routed triage with human review	83% negative recall, ~50% precision. Flag likely-negative reviews, route to the right team via the topic model. Not an autonomous decision-maker.

12. Limitations and Ethics

The corpus is US insurance reviews, not South African medical-aid reviews. Findings transfer by analogy and should be validated on the medical aid's own Hello Peter data. The proxy dataset limits generalisability to South African members.

VADER labels are noisy (Sections 7.2 and 10.3 of the notebook). Sentiment metrics measure agreement with VADER rather than human judgement. A sample of human-labelled reviews would calibrate true performance, which the error analysis suggests is somewhat better than the headline figures.

Negative precision is near 0.50. The model is a triage aid whose flags need human review, not an autonomous decision-maker. The corpus skews positive, so concern themes rest on a minority and the medical aid's actual review mix may differ.

Customer names were removed as personal data. Real deployment should follow POPIA, keep a human in the loop, avoid identifying or penalising individuals from automated sentiment, and not use brand mentions to unfairly target competitors or people.

13. Conclusion

The analysis answers both client questions. The areas of concern reduce to eight review themes, and policy, contact and communication problems carry by far the highest dissatisfaction at 23.9 percent negative and the largest absolute volume of negative reviews. Overall sentiment is predominantly positive: VADER labelled about 88.6 percent of reviews positive against 6.1 percent negative, and the tuned classifier recovers 83 percent of the negative class at macro-F1 0.798. The held-out test macro-F1 sits in line with the cross-validated macro-F1, so there is no sign of overfitting. Because the model measures agreement with VADER-derived labels rather than human truth, it should run as a topic-routed triage tool with human review, not as an automated decision-maker, and its accuracy should be confirmed on a human-labelled medical-aid sample.

14. References

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Code Attribution

All code was written for this project using the documented public APIs of the libraries listed above. No external code was copied. Where a standard community pattern was followed (such as computing topic coherence for a scikit-learn LDA model with gensim's CoherenceModel), it uses each library's documented interface. The VADER compound-score thresholds used to derive the three sentiment classes follow Hutto and Gilbert (2014).